Delta, theta, gamma, and vega explained in plain English with real examples.

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The Greeks are measurements that quantify how an option price will change in response to different market conditions. You do not need to calculate these numbers yourself. You need to understand what each one means so you can make better decisions about which options to buy, when to hold them, and when to exit.

DELTA

How Much the Option Moves Per Dollar of Stock Movement

Delta measures how much your option price changes for every $1 move in the underlying stock. A call with delta 0.50 gains $0.50 for every $1 the stock rises. At-the-money options have delta approximately 0.50. Deep in-the-money options approach 1.0 — they move dollar-for-dollar with the stock. Deep out-of-the-money options approach 0 — barely moving on small stock moves.

THETA

Daily Time Decay — The Cost of Holding an Option

Theta is the amount your option loses in value each calendar day from time passing, all else equal. If an option has theta of -0.05, it loses approximately $5 per contract per day just from the clock ticking — even if the stock does not move at all. Time decay accelerates significantly as expiration approaches. An option with 30 days to expiration decays slowly. An option with 3 days decays rapidly.

GAMMA

The Rate at Which Delta Changes

Gamma measures how quickly delta changes as the stock moves. A high gamma means delta is changing rapidly — the option is becoming more or less sensitive to stock moves very quickly. Gamma is highest for at-the-money options near expiration. This is why 0DTE options are so explosive — a small stock move rapidly changes delta from 0.30 to 0.80. The same dynamic creates massive losses when the stock moves the wrong way.

VEGA

Sensitivity to Changes in Implied Volatility

Vega measures how much your option changes when implied volatility changes by 1%. If vega is 0.10 and IV increases 1%, the option gains $10 per contract even without stock movement. This is why buying options right before earnings — when IV is extremely high — is dangerous. When the company reports and IV collapses, your option can lose value even if the stock moved in your direction.

GREEKS QUICK REFERENCE — IMPACT ON LONG OPTION BUYERS

GREEK	WHAT IT MEASURES	HELPS OR HURTS BUYERS	KEY INSIGHT
Delta	Option move per $1 stock move	Helps — positive delta means profit when stock rises	Higher delta = more participation but more expensive premium
Theta	Daily time decay	Hurts — value lost every day regardless of stock movement	The longer you hold, the more theta costs you
Gamma	Rate delta changes	Helps if moving your way, accelerates losses if not	Near expiration, gamma is highest — moves amplified both ways
Vega	Sensitivity to IV changes	Helps when IV rises after purchase, hurts when IV falls	Buying high-IV options is expensive — IV crush can destroy value

HOW GREEKS CHANGE BASED ON OPTION POSITION AND TIMING

SCENARIO	DELTA	GAMMA	THETA DAMAGE	VEGA SENSITIVITY
Deep In the Money	0.80 to 1.00	Low	Low	Low
Slightly In the Money	0.55 to 0.80	Moderate	Moderate	Moderate
At the Money	~0.50	Highest	Highest	Highest
Slightly Out of Money	0.20 to 0.45	Moderate	Moderate	Moderate
Deep Out of Money	0.01 to 0.20	Low	Low	Low
30+ days to expiration	Normal	Moderate	Slow daily decay	High
7 to 14 days to expiration	Normal	High	Moderate decay	Moderate
1 to 3 days to expiration	Extreme sensitivity	Highest	Severe decay	Low

> **TST CORE PRINCIPLE**
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> The most dangerous combination for options buyers: deep out-of-the-money options with less than a week to expiration purchased when implied volatility is elevated. You need a massive move in a short time while paying a premium that simultaneously collapses when volatility drops. This describes most 0DTE lottery ticket trades.

Full options curriculum including real trade examples in the TST Academy Options section at topstocktrading.com